



BRB CAPITAL GROUP · NGX ANALYST PLATFORM

# Production Deployment Manual

Live-server deployment for the Engineering Department

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Next.js 14 · Node 20 · embedded PGLite · single-instance, persistent-disk deployment

**Read first — the defining constraint.** This application embeds its database (**PGLite**, Postgres compiled to WebAssembly) with on-disk persistence in a local directory. It is therefore a **single-instance** application requiring a **persistent writable volume** and a **long-running Node server**. It is NOT compatible with serverless platforms (Lambda / Fargate-ephemeral), read-only filesystems, or horizontal autoscaling as-is. To scale out later, migrate the data layer to a managed Postgres (\$10). A single always-on Linux host (VM or VPS) + TLS + a durable volume is the correct target.

## 1 Architecture at a glance

- **Framework:** Next.js 14 (App Router) + React 18 + TypeScript, run as a long-lived Node server (next start).
- **Data:** PGLite embedded Postgres persisted to a directory (default `.data/pg`, override with `PGLITE_DIR`). Repositories use plain parameterised SQL.
- **Market data:** all NGN Market API calls are proxied server-side (the API key never reaches the browser), fronted by an in-memory TTL cache to conserve quota.
- **AI features:** Claude API (Anthropic SDK) powers the daily briefing, filing extraction and portfolio review. Keys resolve DB-first (encrypted) then environment.
- **Background worker:** an in-process alert evaluator starts via Next's instrumentation hook (Node runtime only) — another reason a persistent server is required.
- **Auth:** email/password (scrypt), httpOnly cookie sessions, RBAC (analyst < pm < admin). Edge middleware gates pages; every API route self-guards.

## 2 Prerequisites

| Requirement      | Detail                                                                                                                                              |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Node.js          | v20 LTS or newer (built & run on Node 20). Includes npm.                                                                                            |
| Host             | One always-on Linux VM/VPS (e.g. Ubuntu 22.04) with a persistent disk. ~1 vCPU / 1–2 GB RAM suffices for a single firm; size up for build headroom. |
| Reverse proxy    | nginx (or Caddy/Traefik) terminating <b>TLS</b> — mandatory, because the session cookie is issued Secure in production.                             |
| Outbound network | HTTPS egress to <code>api.ngnmarket.com</code> , <code>doclib.ngxgroup.com</code> (filing PDFs) and the Anthropic API.                              |
| NGN Market key   | A live key ( <code>ngm_live_...</code> ). The server's egress IP may need allow-listing in the NGN Market dashboard.                                |

## 3 Environment variables

Provide these to the Node process via your process manager or a root-owned env file (never commit them).

**Required** values must be set before first boot.

| Variable           | Req | Purpose                                                                                                                                                                                                   |
|--------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NODE_ENV           | Yes | Set to <code>production</code> — enables Secure cookies + the encryption-key safety check.                                                                                                                |
| APP_ENCRYPTION_KEY | Yes | Strong random secret ( <code>openssl rand -hex 32</code> ) deriving the AES-256-GCM key that encrypts stored API keys. <b>Must be stable</b> — changing it makes previously saved in-app keys unreadable. |

| Variable          | Req  | Purpose                                                                                                         |
|-------------------|------|-----------------------------------------------------------------------------------------------------------------|
| BRB_SEED_PASSWORD | Yes  | Password for the three seeded staff accounts on first boot. Strong value; rotate/remove accounts after go-live. |
| NGNMARKET_API_KEY | Yes* | NGN Market key. *Or enter it (encrypted) in Admin → Settings; set in env for unattended boot.                   |
| ANTHROPIC_API_KEY | Yes* | Claude key for AI features. Env keys are used directly (never encrypted) — the robust place for it.             |
| PGLITE_DIR        | Rec  | Absolute path to the DB directory on the persistent volume, e.g. /var/lib/brb/pg.                               |
| PORT              | Opt  | Server port (default 3000). Bind the reverse proxy to it.                                                       |

**Encryption-key discipline.** If APP\_ENCRYPTION\_KEY is unset the app derives a key from BRB\_SEED\_PASSWORD — so changing the seed password later silently corrupts stored keys (symptom: could not decrypt ...; treating as unset and AI/keys stop). Set a dedicated APP\_ENCRYPTION\_KEY once and keep it — and BRB\_SEED\_PASSWORD — stable.

## 4 Build & run

### Base host + persistent volume

```
sudo apt update && sudo apt -y upgrade
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt -y install nodejs nginx
node -v                                # expect v20.x

# mount the persistent data disk (skip mkfs on an EXISTING volume!)
sudo mkfs -t ext4 /dev/nvme1n1          # only on a brand-new volume
sudo mkdir -p /var/lib/brb
echo '/dev/nvme1n1 /var/lib/brb ext4 defaults,nofail 0 2' | sudo tee -a /etc/fstab
sudo mount -a
sudo useradd --system --create-home --shell /usr/sbin/nologin brb
sudo mkdir -p /var/lib/brb/pg && sudo chown -R brb:brb /var/lib/brb
```

### Deploy the application

```
sudo mkdir -p /opt/brb-analyst && sudo chown brb:brb /opt/brb-analyst
sudo -u brb git clone <repo-url> /opt/brb-analyst # fresh source only
cd /opt/brb-analyst
sudo -u brb npm ci
sudo -u brb npm run typecheck && sudo -u brb npm test # optional gates
sudo -u brb npm run build
```

Deploy **fresh source** — never copy a development `.data/pg` database or `.env.local`. The live DB seeds itself on first boot.

### Secrets & environment

```
sudo mkdir -p /etc/brb
sudo cp /opt/brb-analyst/.env.production.example /etc/brb/brb.env
sudo nano /etc/brb/brb.env          # fill EVERY required value
openssl rand -hex 32                # -> APP_ENCRYPTION_KEY (set once, never change)
openssl rand -base64 24              # -> BRB_SEED_PASSWORD
sudo chown root:brb /etc/brb/brb.env && sudo chmod 640 /etc/brb/brb.env
```

### Run under systemd

```
sudo cp /opt/brb-analyst/deploy/brb-analyst.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable --now brb-analyst
journalctl -u brb-analyst -f          # watch for '[alerts] background worker started'
curl -s localhost:3000/api/health    # {"ok":true,"status":"healthy",...}
```

(PM2 is an equally valid supervisor: `pm2 start "npm run start" --name brb-analyst`.)

### nginx + TLS

```
sudo cp /opt/brb-analyst/deploy/nginx-brb-analyst.conf /etc/nginx/sites-available/brb-analyst
sudo nano /etc/nginx/sites-available/brb-analyst # set server_name
sudo ln -s /etc/nginx/sites-available/brb-analyst /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default && sudo nginx -t
sudo apt -y install certbot python3-certbot-nginx
sudo certbot --nginx -d analyst.yourfirm.example # cert + auto-renew
sudo systemctl reload nginx
```

TLS is **mandatory**. Without it (and X-Forwarded-Proto=https from the proxy) the Secure session cookie is dropped and sign-in silently fails.

## 5 First-boot hardening

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On first start the app creates its schema and seeds three accounts using `BRB_SEED_PASSWORD`: `analyst@brb.local`, `pm@brb.local`, `admin@brb.local`.

- Sign in over **HTTPS** as `admin@brb.local`, go to **Admin** → **Settings** → **Users & roles**.
- Create real staff accounts (Admin, PFM, Analyst) with strong passwords.
- Delete the seeded `@brb.local` accounts (create your real admin first — the last admin cannot be removed).
- Confirm market data loads (NGN key + IP allow-list) and generate one AI briefing (Claude key).
- Locked out? Stop the service and run `RESET_EMAIL=... RESET_PASSWORD='...' RESET_ROLE=admin npm run reset-admin`, then restart.

## 6 Data persistence & backups

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- All state (users, portfolios, ledgers, AI outputs, encrypted settings) lives in `PGLITE_DIR`. **This directory is the entire database — back it up.**
- Mount it on a persistent, snapshotted volume separate from the app code.
- Back up by snapshotting the volume (or copying the directory during a quiet window). Restore by replacing the directory and restarting.
- Keep a secure record of `APP_ENCRYPTION_KEY` — a restored volume is undecryptable for its stored secrets without the same key.

## 7 Security checklist

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- `NODE_ENV=production` and TLS enforced end-to-end.
- `APP_ENCRYPTION_KEY` strong & stable, from a secrets manager.
- Seeded demo accounts removed; `BRB_SEED_PASSWORD` strong, not reused.
- Firewall: inbound 443 (and 80 redirect/ACME) open; 22 from admin ranges only. Egress IP allow-listed with NGN Market.
- Secrets never in the repo, image, or logs; env file root-owned `chmod 640`.
- Every API route self-guards (middleware excludes `/api`); the NGN proxy requires a signed-in user and restricts account/`*` to admins.
- Password resets revoke sessions; the last admin cannot be demoted/deleted.

## 8 Observability

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- **Health check:** point the load balancer/monitor at `GET /api/health` — 200 when the DB answers, 503 otherwise. Unauthenticated paths are limited to `/api/health`, `/login`, `/brb-logo.png` and static assets.
- **Logs:** the Node process logs to stdout/stderr — capture via `journalctl -u brb-analyst` (or PM2) and ship to your log stack.
- **Quota & audit:** Admin → Quota & Audit shows NGN Market usage and the analyst audit trail.
- **Alert worker:** confirm on boot that the instrumentation worker started (logged once).

## 9 Upgrades & rollback

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```
# CI: npm run typecheck && npm test && npm run build (before promoting)
# snapshot the data volume first, then:
cd /opt/brb-analyst && sudo -u brb git pull && sudo -u brb npm ci && sudo -u brb npm run build
sudo systemctl restart brb-analyst && curl -s localhost:3000/api/health
```

Schema changes are additive/idempotent (`CREATE TABLE / ALTER ... ADD COLUMN IF NOT EXISTS`) and apply on boot — no manual migration. Roll back by checking out the prior release and, if needed, restoring the volume snapshot.

## 10 Scaling beyond one instance

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When one node is no longer enough, migrate off the embedded DB: stand up a managed Postgres, repoint the repository layer's `query()` (in `src/lib/db/client.ts`) at a connection pool — the SQL is standard and backend-agnostic — and export/import the data. The app can then run as multiple stateless replicas behind a load balancer, with the alert worker moved to a single dedicated instance to avoid duplicate evaluation.

## 11 Troubleshooting

| Symptom                                                              | Likely cause & fix                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Server exits on boot with an APP_ENCRYPTION_KEY error                | Working as designed — set a strong APP_ENCRYPTION_KEY in production.                                                                                                                                                                                                 |
| Login succeeds then immediately logs out                             | TLS not terminated / X-Forwarded-Proto not set → Secure cookie dropped. Fix proxy headers and serve over HTTPS.                                                                                                                                                      |
| Dashboard tiles show “API key not configured”                        | Set NGNMARKET_API_KEY (env) or enter it in Admin → Settings; confirm the egress IP is allow-listed.                                                                                                                                                                  |
| AI features / a saved key silently stop; logs show could not decrypt | The in-app key was encrypted under a different master key (i.e. APP_ENCRYPTION_KEY/BRB_SEED_PASSWORD changed). Restore the original key, re-enter the key in Admin → Settings, or set the plain env var (env keys aren't encrypted). Keep the key stable thereafter. |
| No admin can sign in / seed password unknown                         | Stop the service and run <code>npm run reset-admin (\$5)</code> , then restart.                                                                                                                                                                                      |
| Data lost after redeploy                                             | DB directory was not on a persistent volume — mount PGLITE_DIR on durable storage.                                                                                                                                                                                   |
| Filing extraction times out                                          | Raise <code>proxy_read_timeout</code> and <code>client_max_body_size</code> at the proxy.                                                                                                                                                                            |

## 12 Go-live checklist

- Persistent volume mounted; PGLITE\_DIR points to it
- Firewall: 443 open, 80 open (redirect/ACME), 22 admin-IP only; egress IP allow-listed with NGN Market
- `brb.env` complete; APP\_ENCRYPTION\_KEY generated & recorded (stable); BRB\_SEED\_PASSWORD strong; file `chmod 640`
- `npm run build` succeeded; service enabled and active (running)
- `curl localhost:3000/api/health → 200`
- TLS live (certbot); HTTP→HTTPS redirect works; X-Forwarded-Proto set
- Signed in over HTTPS (cookie persists); market data + one AI briefing confirmed
- Seeded `@brb.local` accounts removed; real staff created
- Volume snapshot schedule enabled; APP\_ENCRYPTION\_KEY backed up securely
- Health check wired to `/api/health`; logs shipping

Companion files in the repository: `.env.production.example`, `deploy/brb-analyst.service`, `deploy/nginx-brb-analyst.conf`, `docs/aws-go-live-runbook.md` (AWS EC2 specifics).